Volumes of Revolution

The disk and washer methods

1 Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.

2 The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

The shell method

3 The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.

4 The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.